Credit Constraints, Technology Upgrading, and the Environment

Dana C. Andersen

Abstract: This paper develops a tractable general equilibrium model to analyze the effect of credit constraints on production-generated pollution emissions. The model demonstrates that reducing credit constraints increases the scale of production (scale effect) and increases the number of firms taking up production (market-size effect), while it also reduces emissions per unit of output (technique effect) and increases the share of firms investing in the technology upgrade (upgrading-composition effect). Because the former and latter effects are plausibly confounding in nature, the net effect of credit constraints on pollution emissions is an empirical question. This paper demonstrates, using variation in the timing of credit market reforms, that reducing credit constraints significantly improves air pollution. The results are robust using various approaches, including difference in differences (DID) with a rich set of controls, and an alternative DID approach, wherein the time series data are collapsed around credit reforms into a pre- and post-period.

JEL Codes: D24, D53, Q53, Q55

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ACCESS TO CREDIT figures prominently in the analysis of firm investment decisions, and in turn firm performance and economic growth.[1] Credit market imperfections, including imperfect information, among other incentive problems, however,

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1. The literature on credit constraints and firm investment is vast; see Hubbard (1998) for a survey article. Midrigan and Xu (2014) estimate that credit constraints (frictions)

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preclude investment in fixed assets, thereby constraining technology choices. Technology choices include investment in pollution abatement technologies (e.g., catalytic reduction technologies and sulphur dioxide "scrubber" systems) and, more generally, investment in technologies reducing input requirements, including fuel and material requirements in production (e.g., combined cycle gas turbines in electric power generation and continuous casting technologies in steel manufacturing).[2] Because pollution emissions are inextricably linked to technology and abatement choices, credit constraints bear on environmental performance whenever they influence investment decisions. While significant attention has been given to the relationship between credit constraints and economic performance, no studies to date have systematically explored the effect of credit constraints on environmental performance.

The contribution of this paper is twofold. First, I develop a tractable general equilibrium model to analyze the effect of credit constraints on the environment. The model divides the impact on pollution into firm scale and technique effects and general equilibrium upgrading-composition and market-size effects. Due to the confounding nature of the effects, the overall effect of credit constraints on the environment is an empirical question, though the model demonstrates that reducing credit constraints plausibly leads to a net reduction in pollution emissions. Second, this paper empirically demonstrates that credit market reforms, while reducing imperfect information and widening credit markets, significantly improve air pollution (sulphur dioxide and lead concentration) in both developing and developed countries.

To analyze the effect of credit constraints on aggregate pollution emissions, this paper develops a general equilibrium model with heterogeneous firms based on the Dixit and Stiglitz (1977) model of monopolistic competition.[3] Firm heterogeneity accounts for the dispersion of productivities and emissions intensities across firms and partitions the pool of potential firms, determining which firms will take up production and which firms will invest in a technology upgrade. The technology upgrade

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reduce total factor productivity by 40%, while Hennessy and Whited (2007) estimate that credit frictions represent 13% and 25% of financing costs for large and small firms, respectively. Rajan and Zingales (1998), Levine, Loayza, and Beck (2000), and Beck, Levine, and Loayza (2000) demonstrate that financial development positively affects economic growth.

2. In the case of electricity power generation, combined cycles increase thermal efficiency by using the exhaust of gas turbines to power a steam power plant, thereby reducing fossil fuel inputs. Similarly, in the case of steel manufacturing, continuous casting technologies have eliminated the most energy-intensive stages of production.

3. The Dixit and Stiglitz (1977) model has been widely adopted in various economic fields, such as international trade (e.g., Melitz 2003), as well as the trade-and-the-environment literature (Copeland [2011] surveys the literature, including a section on models with monopolistic competition). One limitation of the Dixit and Stiglitz (1977) (constant elasticity) model is that firms' markups are exogenously fixed by the elasticity of substitution across varieties; however, this assumption turns out to be very convenient and ensures that the model remains tractable while retaining generality in respects that are of first-order importance.

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extends firm productivity by a fixed constant and shifts the intensity of emissions in production. The model demonstrates that credit constraints, however, preclude investment in the technology upgrade for a subset of firms that would otherwise find the investment profitable.

Reducing credit constraints widens the subset of firms investing in the technology upgrade, increasing productivity and in turn output among affected firms. While increasing output increases pollution emissions through the "scale effect," increased productivity reduces emissions intensity whenever productivity and emissions intensity are inversely related, as evidenced by several empirical studies (Cole, Elliott, and Shimamoto 2005; Cole, Elliott, and Wu 2008; Shapiro and Walker 2015). This "technique effect" is reinforced whenever the technology upgrade uses a cleaner production process or mix of inputs. Therefore, among affected firms, reducing credit constraints reduces pollution emissions if the technique effect outweighs the scale effect, which is more likely when (i) emissions intensity is declining in productivity, (ii) the upgrade is inherently cleaner, and (iii) the firm faces an inelastic demand curve.[4]

Taking into account general equilibrium effects implies that reducing credit constraints increases ex ante expected profits, thereby encouraging entry and resulting in a greater mass of firms taking up production. Consequently, pollution emissions increase through the "market-size effect." However, reducing credit constraints increases the share of firms investing in the technology upgrade, resulting in a "upgrading-composition effect." Therefore, the partial equilibrium technique effect is reinforced at the macroeconomic level by the upgrading-composition effect. Reducing credit constraints reduces aggregate emissions if the reduction in emissions among the additional firms investing in the technology upgrade outweighs the increase in emissions among the additional firms entering the market.

To resolve the ambiguity pointed out in the theoretical model, this paper assesses the effect of credit market reforms on pollution emissions. Specifically, I exploit variation in the timing of the establishment of a credit bureau registry, which facilitates the exchange of information across lenders and reduces imperfect information, across countries as exogenous variation in credit constraints. The empirical analysis accounts for unobserved fixed effects and aggregate time effects, as well as control variables, to produce a difference in differences (DID) estimate of credit reforms. The results demonstrate that the establishment of a credit bureau significantly reduces sulphur dioxide and lead concentrations, which are important production-generated pollutants, across 305 pollution-monitoring sites in 37 countries. Using the preferred specification, credit reforms reduce sulphur dioxide and lead concentrations by approximately 0.5% and 1.1% after 5 years. Moreover, consistent with empirical studies document-

4. Inelastic demand implies that an increase in productivity results in reduced prices rather than increased output, thereby limiting the scale effect.

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ing that the credit reforms are particularly important for credit intermediation in countries with French legal origin (Djankov, McLiesh, and Shleifer 2007), the empirical analysis demonstrates that the effect of credit reforms on pollution is particularly acute in French legal origin countries. Finally, following the approach suggested by Bertrand, Duflo, and Mullainathan (2004), the results are consistent using an alternative DID approach, wherein the time series data are collapsed into a pre- and post-period and pollution emissions are analyzed around the establishment of a credit bureau.

This paper is related to the empirical literature exploring the relationship between firm environmental performance and broadly defined financial performance. Gray and Deily (1996) and Shadbegian and Gray (2005) find that more profitable firms are not more likely to comply with environmental standards, whereas Maynard and Shortle (2001) find that more profitable and less leveraged firms are more likely to invest in a clean technology.[5] Earnhart and Lizal (2006, 2010) report that profits are positively associated with air pollution emissions, whereas value added is negatively associated with air pollution emissions. Earnhart and Segerson (2012) contribute to the literature by pointing out that the correlation, if one exists, between financial performance and environmental performance might be the consequence of correlation between profits and unobservable factors, such as liquidity and solvency risk. Earnhart and Segerson (2012) is the first study to advance a conceptual framework to analyze various dimensions of financial status on compliance with environmental regulations and to conduct an empirical analysis of various dimensions of financial status, finding that liquidity and solvency risks do not in general play an important role in industrial wastewater discharges of chemical manufacturing facilities.

The remainder of this paper is organized as follows. Section 1 presents the theoretical model. Section 2 discusses the data, estimation strategy, results, and robustness checks. Section 3 concludes.

1. MODEL

1.1. Overview

This section presents a general equilibrium model with heterogeneous firms based on the Dixit and Stiglitz (1977) model of monopolistic competition.[6] There are an infinite number of time periods, and in each discrete period, a pool of initially iden-

5. Gray and Deily (1996) analyze 41 steel plants, Shadbegian and Gray (2005) analyze 116 pulp and paper mills, and Maynard and Shortle (2001) analyze 75 bleached kraft pulp mills.

6. Because the link between credit constraints and pollution emissions is general, the model can be adapted to analyze various problems related to endogenous technology decisions and the environment.

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tical potential entrants face a dynamic forward-looking entry decision, where paying a market entry cost confers a draw of a random productivity. Firms with productivity draws conferring present value profits in excess of a fixed production cost will take up production, while firms with lower productivity draws will immediately exit. Firms not exiting will remain in the market in subsequent periods but face an exogenous probability that a random shock will force them to exit. Moreover, there exists a discrete technology upgrade, but due to credit constraints, only a subset of firms are able to finance the investment. With a given technology, firms produce a final good using a variable composite input and generate pollution emissions as a byproduct of production, which depends on the technology employed. Figure 1 provides an overview of these decisions. Note that in the model all relevant decisions are made simultaneously at the beginning of the period.

1.2. Consumers

Utility of a representative consumer exhibits constant elasticity of substitution (CES) with elasticity of substitution $\sigma > 1$

$$U = \left[\int_{v \in V} q(v)^{\frac{\sigma - 1}{\sigma}} dv \right]^{\frac{\sigma}{\sigma - 1}}, \tag{1}$$

where $q(v)$ is demand for variety v, and V is the set of available varieties.[7] Expenditure of the representative consumer is denoted Y and the price of variety v is denoted $p(v)$. Utility maximization, subject to the budget constraint $\int_{v \in V} p(v)q(v)dv = Y$, yields the isoelastic demand function for variety v

$$q(v) = YP^{\sigma - 1} p(v)^{-\sigma}, \tag{2}$$

where P is the CES price index.

1.3. Producers

There is a continuum of firms, each producing a unique variety. Production is linear in a composite input (labeled "labor" for convenience), which includes human capital, physical capital, fuel, energy, materials, and so on. The labor market is perfectly competitive, with an equilibrium wage rate w. Profit maximization for firms treating wages as given implies that marginal production costs are w/ϕ, where ϕ is the firm's

7. Because the model abstracts from endogenous pollution policy responses, I do not consider disutility associated with environmental damage. However, incorporating environmental damage using a weakly separable utility function (e.g., $W(Z, U)$ where U is given by equation [1] and Z is aggregate pollution) would not change the analysis as the marginal rate of substitution (and thus the elasticity of substitution) of two varieties would be independent of pollution.

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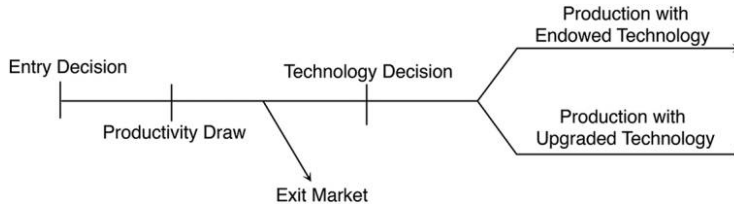

Figure 1. Overview of firm decisions

productivity parameter. Access to credit, as explicated below, determines if the firm adopts a technology upgrade, which increases productivity by a fixed scalar $\delta > 0$. In this case, marginal production costs are $w/[\phi(1+\delta)]$. As in Bustos (2011), the technology upgrade can be interpreted, in general, to represent firm actions to reduce the marginal cost of production or to increase the quality of the product.

Bustos (2011) and Midrigan and Xu (2014) are two recent examples of models incorporating endogenous technology decisions using an identical productivity-scaling factor. That credit constraints reduce firm productivity and growth is supported by a number of empirical studies (Cooley and Quadrini 2001; Albuquerque and Hopenhayn 2004; Clementi and Hopenhayn 2006). Moreover, Midrigan and Xu (2014) and Cole, Greenwood, and Sanchez (2015) demonstrate that credit constraints lead to productivity losses through low levels of technology adoption in particular. In a model with an endogenous technology upgrade subject to credit constraints, Midrigan and Xu (2014) estimate that the relative efficiency of the more productive technology adoption is 0.27, which is directly comparable to the parameter δ.

For notation, firms not adopting the more productive technology are denoted with the superscript c, while firms adopting the technology are denoted with the superscript u. Profit maximization implies that firms set prices equal to a constant markup over marginal cost

$$p^c(v) = \frac{w/\phi}{\rho} \text{ and } p^u(v) = \frac{w/[\phi(1+\delta)]}{\rho}, \tag{3}$$

where $\rho \equiv (\sigma - 1)/\sigma$. Using the derived isoelastic demand function implies that equilibrium output and revenue for firms not investing in the upgrade are given by

$$q^c(\phi) = YP^{\sigma-1}\left(\frac{w}{\phi\rho}\right)^{-\sigma} \text{ and } r^c(\phi) = YP^{\sigma-1}\left(\frac{w}{\phi\rho}\right)^{1-\sigma}. \tag{4}$$

Moreover, firms adopting the technology upgrade set lower prices, produce more, and generate more revenue. That is,

$$\frac{q^u(\phi)}{q^c(\phi)} = (1+\delta)^{\sigma} > 1 \text{ and } \frac{r^u(\phi)}{r^c(\phi)} = (1+\delta)^{\sigma-1} > 1. \tag{5}$$

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Pollution is emitted as a joint output of production and is equal to emissions intensity (emissions per unit of output) times output. I assume that emissions intensities are given by

$$e^c(\phi) = \frac{1}{\phi^\alpha} \text{ and } e^u(\phi) = \frac{\gamma}{(\phi(1+\delta))^\alpha}. \tag{6}$$

Emissions intensity therefore changes monotonically with firm productivity, with the technology parameter α determining the extent to which this is the case. Because the model abstracts from endogenous environmental regulations, the emissions intensity of firms not upgrading can be considered normalized to unity, while the parameter γ reflects the difference in emissions intensity associated with the upgrade that are not due to differences in productivity.[8]

What links technology upgrading and emissions intensity? First, firms invest in the technology upgrade to reduce marginal production costs, which corresponds to a reduction in input intensity. If the technology upgrade is factor neutral (e.g., Cobb-Douglas) then the intensity of all inputs, including fuels and materials, would be reduced, thereby reducing pollution emissions even in the absence of environmental regulations. Moreover, to the extent that pollution emissions are regulated, upgrading to a technology that (all else constant) generates less emissions, including adopting clean technologies and installing pollution abatement equipment, would also reduce production costs.[9] Second, there might be differences in emissions intensity not due to, at least in part, differences in productivity. For example, the technology upgrade might entail a different production process and mix of inputs, including switching fuel types or materials in production.[10] Moreover, because the product quality might be upgraded, the production process would likely be altered to produce the higher quality product.[11] Distinguishing between the former (α)

8. As discussed at the end of this section, to the extent that credit constraints reduce income and environmental quality is a normal good, incorporating endogenous environmental regulations would reinforce the effect of reducing credit constraints.

9. Maynard and Shortle (2001) find that firms with a higher cost of capital are less likely to adopt cleaner technologies in the kraft pulp industry. Moreover, Gray and Shadbegian (1998) find that productivity investment projects are often coupled with investments in pollution abatement.

10. More specifically, an electricity producer upgrading to a combined cycle generator might use a cleaner mix of fossil fuels, such as more natural gas and less coal, thereby further reducing emissions intensity.

11. For example, there are thousands of different grades of steel that generate a range of environmental impacts (https://worldsteel.org).

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and the latter (γ) effects is not inconsequential, as the former would involve a scale effect while the latter would not.

From a theoretical point of view, expression (6) is a generalization of the Cobb-Douglas pollution-production technology adopted by Copeland and Taylor (2003), in which α can be interpreted as the cost share of nonpollution inputs (Shapiro and Walker 2015). Shapiro and Walker (2015) estimate this parameter, finding that the elasticity of sulphur dioxide emissions (among other pollutants) with respect to total factor productivity is approximately -1, suggesting that α is approximately 1. However, abstracting from the Cobb-Douglas pollution-production formulation, this finding can be interpreted as (i) more productive firms use less inputs in general, (ii) more productive firms use less dirty inputs and production processes, or (iii) more productive firms engage in more pollution abatement. This study, as well as others documenting that productivity is inversely related to pollution (Cole et al. 2005, 2008), therefore suggests that either emissions intensity is decreasing in productivity ($\alpha > 0$) or there are differences in emissions intensity not due to differences in productivity but that are correlated with productivity ($\gamma < 1$), or both.

For simplicity and clarity, it is expedient to distinguish between output and emissions associated with the endowed technology and "additional" output and emissions associated with the technology upgrade, which I denote using the superscript a. For example, additional output is given by $q^a = q^u - q^c$. I will refer to variables corresponding to the endowed technology production as "constrained" values, while variables corresponding to the additional values associated with the technology adoption as "additional" values.

It follows that additional output and revenue are given by

$$q^a(\phi) = ((1+\delta)^\sigma - 1)q^c(\phi) \text{ and } r^a(\phi) = \Delta r^c(\phi), \tag{7}$$

where $\Delta \equiv \left((1+\delta)^{\sigma-1} - 1\right) > 0$. As expected, additional output and revenue are proportional to output and revenue generated by the endowed technology. Similarly, pollution emissions are given by

$$z^c(\phi) = \frac{q^c(\phi)}{\phi^\alpha} \text{ and } z^a(\phi) = (\Gamma - 1)z^c(\phi), \tag{8}$$

where $\Gamma \equiv \gamma(1+\delta)^{\sigma-\alpha} > 0$.

Lemma 1: Additional pollution emissions z^a is negative if and only if $\Gamma < 1$. In this instance, I will refer to the technology upgrade as "pollution decreasing," otherwise I will refer to the technology upgrade as "pollution increasing."

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The technology upgrade is therefore more likely pollution decreasing whenever (i) the upgrade is inherently cleaner, (ii) the productivity gain is modest (assuming $\alpha < \sigma$), (iii) emissions intensity is declining in productivity, and (iv) the firm faces a more inelastic demand curve.

Relying on previous studies, it is possible to shed light on some plausible parameter values: (i) Shapiro and Walker (2015) estimate an elasticity of substitution around 4; (ii) Midrigan and Xu (2014) estimate a relative efficiency of technology adoption around 0.27; and (iii) Shapiro and Walker (2015) estimate an elasticity of pollution with respect to productivity around 1. It should be noted that there is not a general consensus regarding the second and third parameter values. Moreover, as discussed, a negative correlation between emissions intensity and productivity is consistent with both $\alpha > 0$ or $\gamma < 1$.

Figure 2 plots ($\delta - \gamma$) pairs such that the technology upgrade is pollution increasing and pollution decreasing using an elasticity of substitution $\sigma = 4$ and various values of $\alpha = 0$, 1, and σ. The graph highlights that even after restricting attention to plausible parameter values, both pollution-increasing and pollution-decreasing upgrades are empirically possible.

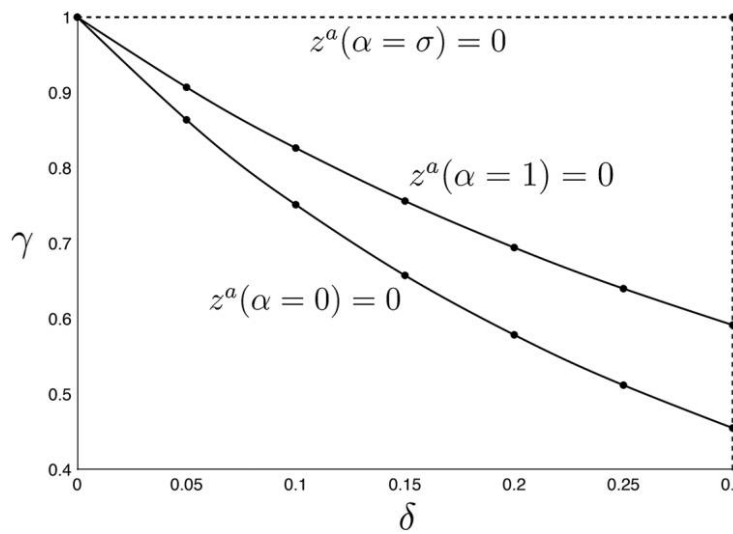

Figure 2. Pollution-decreasing and pollution-increasing upgrades. The graph uses an elasticity of substitution $\sigma = 4$. The area below the curves corresponds to a pollution-decreasing upgrade for $\alpha = 0$, while the area above the curves corresponds to a pollution-increasing upgrade for $\alpha = 1$. If ($\alpha = 0$) then the area between the curves corresponds to the pollution-decreasing (increasing) region as well.

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1.3.1. Fixed and Upgrading Costs

All firms have a fixed production and distribution cost wf, whereas firms adopting the technology upgrade incur an additional fixed investment cost $w\beta$. Constrained and additional profits, defined as corresponding revenue less labor and fixed costs, are given by[12]

$$\pi^c = \frac{r^c}{\sigma} - wf \text{ and } \pi^a = \frac{r^a}{\sigma} - w\beta. \tag{9}$$

Credit constraints imply that upgrading costs cannot exceed a fraction of unconstrained revenue less labor cost. That is, $w\beta \leq \theta r^a/\sigma$, where $0 < \theta < 1$. The assumption that $\theta < 1$ implies the existence of firms that find upgrading profitable but that cannot finance the upgrade. Credit constraints are therefore modeled, using a reduced-form parameter, in the spirit of credit rationing in markets with imperfect information (Stiglitz and Weiss 1981), wherein identifiable groups of borrowers are unable to obtain loans at any interest rate.[13] Note that credit constraints are monotonically decreasing in θ, implying that credit constraints are decreasing in any monotonically increasing function of θ. Consequently, applying the credit constraint condition to constrained, rather than unconstrained, revenue less corresponding labor costs is immaterial as the two are proportional.[14] While the model focuses on credit rationing for clarity, accounting for endogeneous borrowing costs does not qualitatively change the results.[15]

The fixed production and upgrading costs define zero-profit and credit-constraint conditions. In particular,

$$\frac{r^c(\phi^c)}{\sigma} = wf \text{ and } \frac{r^a(\phi^u)}{\sigma} = \frac{w\beta}{\theta}, \tag{10}$$

12. Additional profits are profits with the technology upgrade less profits without the upgrade, where profits with the technology upgrade are given by

$$\pi^u = \frac{r^u}{\sigma} - (wf + w\beta) = \overbrace{\left(\frac{r^c}{\sigma} - wf\right)}^{\pi^c} + \overbrace{\left(\frac{r^a}{\sigma} - w\beta\right)}^{\pi^a}.$$

13. Consistent with Stiglitz and Weiss (1981), if credit rationing exists then borrowers with lower returns to upgrading will be credit rationed. This result is also consistent with empirical studies documenting that small firms are more likely to face financing constraints (Hennessy and Whited 2007).

14. That is, $r^u = \Delta r^c$, implying that we can apply the following monotonic transformation: $\theta^* = \theta/\Delta$.

15. To be precise, incorporating additional borrowing costs to finance fixed investments, which are increasing in credit constraints, reinforces result 1 and does not bear on any of the subsequent results. The details are provided in appendix A1.

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where ϕ^c is the cutoff productivity for producing in the market, while ϕ^u is the cutoff productivity for investing in the technology upgrade. I assume firms that invest in the technology upgrade also find it profitable to produce without the technology upgrade, which follows if and only if $f < \theta\beta\Delta$, implying that $r^c(\phi^c) < r^a(\phi^u) \Leftrightarrow \phi^c < \phi^u$. In other words, all firms that are able to finance the investment in the upgrade find it profitable to produce, but not all firms that find it profitable to produce are able to finance the investment in the upgrade.

1.3.2. Firm Entry and Exit

There is an unbounded pool of potential entrants deciding on paying a fixed market entry cost wf^e, which allows them to draw a random productivity from the common distribution $G(\phi)$. After realizing their productivity, firms then decide whether to start producing and whether to invest in the technology upgrade. Recall that there is an infinite number of time periods, and every period an exogenous negative shock precludes production for a fraction of producing firms η.

I assume that firm productivities are Pareto distributed, with the lower bound normalized to one, $G(\phi) = 1 - \phi^{-k}$ and $g(\phi) = k\phi^{-(k+1)}$.[16] To ensure that average per-firm output is finite, I assume that $k > \sigma$. With the ex ante productivity distribution being Pareto, the ex post productivity distributions of active firms and firms investing in the technology upgrade are Pareto. That is,

$$\mu^c(\phi) = \frac{k}{\phi}\left(\frac{\phi^c}{\phi}\right)^k \text{ and } \mu^u(\phi) = \frac{k}{\phi}\left(\frac{\phi^u}{\phi}\right)^k. \tag{11}$$

Free entry implies that the ex ante present value of expected profits has to be equal to the cost of entering the productivity draw.

$$(\phi^c)^{-k}\frac{\overline{\pi}}{\eta} = wf^e, \tag{12}$$

where $\overline{\pi}$ is average profits of active firms, and $(\phi^c)^{-k}$ is the ex ante probability that the firm produces. The left-hand side is the present value of expected profits from the point of view of a potential entrant that does not yet know its productivity, which is equal to the cost of entry in equilibrium.

1.3.3. Labor Market Clearing

Denote the mass of firms entering the productivity draw as M^e, and the mass of firms taking up production and the mass of firms investing in the technology upgrade as M^c and M^u ($M^c > M^u$), respectively. Full employment in production, as well as market entry, is written as

16. Eaton, Kortum, and Kramarz (2011), among others, document that firm size approximately follows this distribution.

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$$L = M^e f^e + M^c\left(f + \int_{\phi^c}^{\infty} \frac{q^c}{\phi}\mu^c(\phi)d\phi\right) + M^u\left(\beta + \int_{\phi^u}^{\infty} \frac{q^a}{\phi(1+\delta)}\mu^u(\phi)d\phi\right), \quad (13)$$

where L is the exogenous labor supply. In equilibrium, the mass of firms taking up production is equal to the mass of firms stopping production, implying that $(\phi^c)^{-k}M^e = \eta M^c$. Using equations (7), (10), and (12) in the employment condition (13) solves for the equilibrium M^c and M^u.[17] That is,

$$M^c = \frac{L/(\kappa_1\sigma)}{\left(f + \left(\frac{\phi^c}{\phi^u}\right)^k\left(\frac{\beta}{\theta}\right)\right)} \text{ and } \frac{M^u}{M^c} = \left(\frac{\theta\Delta f}{\beta}\right)^{\frac{k}{\sigma-1}}, \quad (14)$$

where $\kappa_1 = k/(1+k-\sigma)$. Thus, the ratio of firms investing in the technology upgrade to all firms is determined by exogenous parameters, including θ, which increases the percentage of firms investing in the technology upgrade.

1.4. Baseline Analysis

This section presents three main results. Result 1 analyzes the impact of reducing credit constraints on the mass of firms taking up production (market-size effect) and the percentage of firms investing in the technology upgrade (upgrading-composition effect). Result 2 analyzes the impact of reducing credit constraints on average constrained and unconstrained output (scale effect) and pollution emissions (technique effects). Result 3 describes necessary and sufficient conditions such that reducing credit constraints reduces aggregate emissions.

Result 1: Upgrading-Composition and Market-Size Effect: Reducing credit constraints increases the percentage of firms investing in the technology upgrade and the mass of firms taking up production. That is,

$$\frac{\partial\left(\frac{M^u}{M^c}\right)/\frac{M^u}{M^c}}{\partial\theta/\theta} = \frac{k}{\sigma-1} > 0 \text{ and } \frac{\partial M^c/M^c}{\partial\theta/\theta} = \frac{\beta/\theta\left(\frac{\phi^c}{\phi^u}\right)^k\left(\frac{k}{\sigma-1}-1\right)}{\left(f + \left(\frac{\phi^c}{\phi^u}\right)^k\left(\frac{\beta}{\theta}\right)\right)} > 0. \quad (15)$$

Proof: Follows from expression (14). QED

Because reducing credit constraints increases the percentage of firms investing in the technology upgrade and the mass of firms taking up production, a corollary of result 1 is that the mass of firms that invest in the technology upgrade must increase

17. The details of the derivation are provided in appendix A1.

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as well. On the other hand, the effect of reducing credit constraints on the mass of firms taking up production but not investing in the technology upgrade is ambiguous.

Corollary 1.1: Productivity Cutoffs: Reducing credit constraints reduces the cutoff productivities for investing in the technology upgrade and taking up production. That is,

$$\frac{\partial \phi^c/\phi^c}{\partial \theta/\theta} = -\frac{1}{k}\frac{\partial M^c/M^c}{\partial \theta/\theta} < 0 \text{ and } \frac{\partial \phi^u/\phi^u}{\partial \theta/\theta} = \frac{\partial \phi^c/\phi^c}{\partial \theta/\theta} - \frac{1}{\sigma - 1} < 0. \tag{16}$$

The corollary follows immediately from the fact that the mass of firms investing in the technology upgrade and taking up production increases if and only if the corresponding productivity threshold decreases.

Define average output and average emissions for firms taking up production as $\overline{q}^c$ and $\overline{z}^c$. That is,

$$\overline{q}^c = \int_{\phi^c}^{\infty} q^c \mu^c(\phi) d\phi \text{ and } \overline{z}^c = \int_{\phi^c}^{\infty} z^c \mu^c(\phi) d\phi. \tag{17}$$

And define $\overline{q}^a$ and $\overline{z}^a$ analogously.

Result 2: Scale and Technique Effects (Average Output and Emissions): Reducing credit constraints reduces average output and pollution emissions among firms taking up production. Moreover, it reduces average additional output and pollution emissions among firms investing in the technology upgrade. That is,

$$\frac{\partial \overline{q}^c/\overline{q}^c}{\partial \theta/\theta} = \frac{\partial \phi^c/\phi^c}{\partial \theta/\theta} < 0 \text{ and } \frac{\partial \overline{z}^c/\overline{z}^c}{\partial \theta/\theta} = (1-\alpha)\frac{\partial \phi^c/\phi^c}{\partial \theta/\theta} < 0 \tag{18}$$

$$\frac{\partial \overline{q}^a/\overline{q}^a}{\partial \theta/\theta} = \frac{\partial \phi^u/\phi^u}{\partial \theta/\theta} - 1 < 0 \text{ and } \frac{\partial \overline{z}^a/\overline{z}^a}{\partial \theta/\theta} = (1-\alpha)\frac{\partial \phi^u/\phi^u}{\partial \theta/\theta} - 1 < 0. \tag{19}$$

Proof: See appendix A1 (all apps. available online). QED

Result 2 is the consequence of general equilibrium effects bearing on the composition of firms. In particular, reducing credit constraints encourages (i) taking up production among firms that otherwise would have exited the market and (ii) investment in the technology upgrade among firms that otherwise would have been credit constrained. In general, these are less productive firms with lower output and in turn lower emissions, though higher emissions intensity.

Result 2 demonstrates that reducing credit constraints decreases average output to a greater extent than it decreases emissions. Thus, a corollary of result 2 is that reducing credit constraints increases average emissions intensity, defined as average emissions per unit of output ($\overline{e}^c = \overline{z}^c/\overline{q}^c$ and $\overline{e}^a = \overline{z}^a/\overline{q}^a$).

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Corollary 2.1: Average Emissions Intensity: Reducing credit constraints increases average emissions intensity. That is,

$$\frac{\partial \bar{e}^c/\bar{e}^c}{\partial\theta/\theta} = -\alpha\frac{\partial\phi^c/\phi^c}{\partial\theta/\theta} > 0 \text{ and } \frac{\partial \bar{e}^a/\bar{e}^a}{\partial\theta/\theta} = -\alpha\frac{\partial\phi^u/\phi^u}{\partial\theta/\theta} > 0. \tag{20}$$

Define the "share" of pollution emissions from firms taking up production and firms investing in the technology upgrade as $s^c = \bar{z}^c M^c/Z$ and $s^u = \bar{z}^a M^u/Z$, where $s^c + s^u = 1$. The impact of reducing credit constraints on aggregate emissions is therefore

$$\frac{\partial Z/Z}{\partial\theta/\theta} = s^c\left(\frac{\partial(\bar{z}^c M^c)/\bar{z}^c M^c}{\partial\theta/\theta}\right) + s^u\left(\frac{\partial(\bar{z}^a M^u)/\bar{z}^a M^u}{\partial\theta/\theta}\right). \tag{21}$$

Thus, $s^u < 0$ alludes to the possibility that reducing credit constraints might lower aggregate pollution emissions.

Result 3: Aggregate Pollution Emissions: The effect of reducing credit constraints on aggregate pollution emissions depends on the magnitude of the market size effect and the magnitude and sign of additional emissions. That is,

$$\frac{\partial Z/Z}{\partial\theta/\theta} = \xi_1\varepsilon^M + \xi_2 s^u, \tag{22}$$

where $\xi_1 = (k + \alpha - 1)/k$, $\xi_2 = (k + \alpha - \sigma)/(\sigma - 1)$, and $\varepsilon^M = [(\partial M^c/M^c)/(\partial\theta/\theta)]$.

Proof: See appendix A1. QED

Corollary 3.1: Reducing credit constraints decreases aggregate pollution emissions if and only if the share of additional pollution emissions is negative ($s^u < 0$) and sufficiently large in magnitude relative to the market size effect. That is,

$$\left|\frac{\varepsilon^M}{s^u}\right| < \frac{\xi_2}{\xi_1} \Leftrightarrow \frac{\partial Z/Z}{\partial\theta/\theta} < 0. \tag{23}$$

Because the statement is an if and only if statement, reducing credit constraints increases aggregate emissions whenever the complement statement is true.

1.5. Discussion: From Theory to Estimation

The model divides the effect of credit constraints on aggregate emissions into technique, scale, upgrading composition, and market size. Reducing credit constraints reduces aggregate emissions if and only if the reduction in pollution emissions resulting in firms investing in the technology upgrade outweighs the increase in the mass of firms taking up production from a reduction in credit constraints. Because the average reduction in pollution emissions among firms investing in the technology is pro-

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portional to $\Gamma - 1$, which can be positive or negative according to lemma 1, the net effect of credit constraints on pollution emissions is ambiguous. Moreover, while it might be possible to estimate the effect of reducing credit constraints on firm entry, it is not possible to estimate the share of emissions generated by unconstrained firms (s^u) as well as other parameters, ruling out the possibility of estimating structural parameters. The model, however, yields a relatively simple reduced-form relationship between credit constraints and aggregate pollution emissions ($\xi_1 \varepsilon^M + \xi_2 s^u$). The aim of the empirical analysis is to determine the sign of this reduced-form parameter, revealing the net effect of credit constraints on aggregate pollution emissions.

Because the model imposes several simplifying assumptions to gain tractability, several generalizations are necessary to bridge the gap between the theory and empirics. For example, the model focuses on a steady-state equilibrium in which various parameters are fixed over time, such as the supply of the composite input and the stringency of environmental regulations. Because the empirical analysis relies on panel data with unobserved fixed effects, generalizing the empirical model to account for time-varying factors such as factor accumulation, which would affect pollution through scale effects, and environmental policy reforms, which would affect pollution emissions through technique and scale effects, will be necessary. Moreover, factor accumulation might not be factor neutral, implying that the composition of the composite input would not be fixed over time, and the composition of inputs might bear on pollution emissions (Antweiler et al. 2001). Therefore, the empirical model should account for a broader set of controls, including both physical and human capital.

To the extent that these additional factors are endogenous to the model, and are related to credit constraints in particular, generalizing the empirical model to account for a broader set of controls bears on the interpretation of the relationship between credit constraints and pollution emissions. Because incorporating endogenous environmental regulations reinforces the effect of credit constraints, accounting for environmental regulations implies that the effect of credit constraints would be underestimated. The implication of accounting for the composition of inputs depends on the relationship between various inputs and pollution, which is an empirical question. Therefore, these issues are discussed in the subsequent section after presenting the primary empirical results (section 2.3.2).

2. EMPIRICAL ANALYSIS

Economy-wide credit market imperfections result in a subset of credit-constrained firms that cannot invest in potentially profitable technology upgrades. An empirical counterpart to the model would therefore be a measure of firms not financing profitable investments due to various credit market imperfections. An ideal analysis of the relationship between credit constraints and aggregate pollution emissions would therefore entail random assignment of this measure across countries. In this case, comparing variation in pollution emissions across countries receiving more or less credit

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constraints would identify the causal effect on pollution emissions. Of course, this experiment is impossible in practice.

This paper exploits institutional reforms occurring at different points in time across countries. Specifically, I use the introduction of a credit bureau registry, which facilitates the exchange of credit information across lenders. The establishment of a credit bureau therefore is associated with a pronounced reduction in asymmetric information, which is a primary cause of credit market imperfections. Moreover, empirical studies demonstrate that the introduction of a credit registry facilitates credit intermediation, and this paper corroborates that the reform increases borrowing.[18]

The empirical analysis uses both panel regression analysis and an alternative DID approach to analyze the effect of credit constraints on pollution emissions. The dependent variable is pollution concentration at a pollution-monitoring site each year, where there are multiple sites within a country. In the panel regression analysis, identification takes advantage of within-country variation in credit reforms, and I introduce site and year effects, as well as time-varying country covariates. The alternative DID approach looks at changes in pollution concentration around the credit reforms. Following the recommendation by Bertrand et al. (2004), I restrict the sample to countries that undertook the reform during the period of study (and have sufficient observations before and after the reform) and run a cross-section regression using average pollution emissions before and after the reform.

2.1. Data Sources

This paper uses data on private credit and credit institutional reforms from Djankov et al. (2007). Air pollution data are from the World Health Organization (WHO) Automated Meteorological Information System (AMIS), provided by the US Environmental Protection Agency (EPA). Country covariates are from the World Bank's World Development Indicators (WDI), human capital measures are from Barro and Lee (2013), and physical capital is from Amadou (2011). Other data sources are discussed throughout the text.

2.1.1. Pollution

Because the theoretical model is relevant for production-generated pollution, I focus on pollution emitted as a by-product of production, rather than consumption-generated pollution. The Global Environment Monitoring Systems (GEMS) was established to monitor the concentrations of various pollutants using comparable measuring devices, thereby producing consistent measures of pollution across developing and developed countries. While there are many production-generated pollutants,

18. Djankov et al. (2007) review the empirical literature documenting that credit bureaus facilitate lending.

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among the pollutants with consistent data in the GEMS database (sulfur dioxide, lead, volatile organic compounds, and carbon monoxide), sulfur dioxide and lead are the only two pollutants with the largest share of emissions from production-generated sources.[19] Sulfur dioxide and lead are two common production-generated air pollutants that are suitable proxies for the impact of industrial activities on environmental quality overall.

Sulfur dioxide and lead pollution can be attributed to anthropogenic sources, primarily industrial processes and electricity generation, and entail significant impacts to human health. While sulphur dioxide (SO_2) is a naturally occurring gas emanating from sources such as volcanoes, sea spray, and decaying organic matter, anthropogenic sources are responsible for between one-third and one-half of total concentration (Kraushaar and Ristinen 1993). On the other hand, lead is a naturally occurring mineral found in ores in the earth's crust, while anthropogenic sources are responsible for practically all of the total concentration found in the air. According to the US EPA, sulphur dioxide emissions are generated from burning fossil fuels, typically at power plants (73%) and other industrial facilities (20%).[20] The primary sources of lead are ore and metal processing, and lead concentrations are highest near lead smelters. Exposure to both types of pollution causes adverse health effects: SO_2 is linked to a number of effects on the respiratory system, while lead is toxic to many organs and tissues and is particularly toxic to children.

While GEMS has been recording pollution concentrations starting in the early 1970s, WHO reports that data comparability may be limited as monitoring techniques and procedures were modified to ensure consistency. Following the recommendations of WHO, I use the most consistent monitoring data starting in 1987 and ending in 1999, at which point GEMS discontinued pollution monitoring. In general, GEMS sites monitor sulfur dioxide, lead, or both; and the data comprise summary statistics for the yearly distributions of concentrations at each site.

Although WHO requests that all member states report pollution data, several countries lack air quality monitoring networks or the ability to process and report

19. López, Galinato, and Islam (2011) calculate the share of pollution generated from (i) production (e.g., industrial processes and electricity generation), (ii) consumption (e.g., road vehicles and residential wood combustion) or (iii) "both" (e.g., fuel combustion, non-road equipment, solvent use, waste disposal, and miscellaneous). Because the both category includes sources that can be production or consumption, it is not possible to attribute it to one or the other. The shares of SO_2 are (i) 80%, (ii) 2%, and (iii) 18%, whereas lead is (i) 56%, (ii) 0%, (iii) and 44%.

20. All data and statistics can be found at the EPA's website under the six "criteria pollutants" (http://epa.gov/airquality/urbanair).

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data satisfying data quality standards. The sulfur dioxide data set consists of 2,252 observations from 305 sites located in 37 countries, while the lead data set consists of 783 observations from 137 sites located in 31 countries.[21] The sulfur dioxide sample therefore encompasses a larger set of countries and greater number of observations per country. Because several sites start recording pollution concentrations after the first year in the panel, and several sites stop recording pollution concentrations before the last year, selection bias is a potential concern, which is discussed in section 2.4.

2.1.2. Credit Constraints

As mentioned, an ideal measure of credit constraints would be a measure of firms not financing profitable investments due to various credit market imperfections. While numerous studies have advanced methodologies for determining whether particular firms are credit constrained, such measures are not available at the country level.[22] In selecting a proxy for credit constraints, this paper turns to economic theory on the determinants of credit constraints and empirical studies analyzing their importance in credit availability. Economic theory suggests that adverse selection in credit markets arises when lenders are unable to observe characteristics of borrowers, such as the riskiness or viability of investments (Jaffee and Russell 1976; Stiglitz and Weiss 1981; among many others). Imperfect information therefore results in credit rationing or high borrowing rates (or both), precluding potentially profitable investments. Conversely, reducing imperfect information mitigates credit rationing as lenders will be more willing to extend credit whenever they know more about potential borrowers, including past credit history and current indebtedness.

Credit bureau registries facilitate the exchange of information across lenders concerning the creditworthiness of existing or potential borrowers. In some cases they are owned and operated by the lenders themselves and in other cases by an independent third party for profit. Insofar as lenders provide their private data to the credit registry, lenders are granted access to the common database, with various sanctions, ranging from fines to loss of membership, used to deter free riding (Jappelli and Pagano 2002). The theoretical literature advances three effects of credit bureaus (Jappelli and Pagano 2002). First, credit bureaus reduce asymmetric information, improving the lender's knowledge of credit applicants and permitting a more accurate prediction of the probability of repayment. Second, credit bureaus level the playing field within the credit market, thereby reducing informational rents and promot-

21. Table B1 in appendix B lists the set of countries in the SO_2 and lead samples.

22. Moreover, various approaches for measuring credit constraints, such as surveys and indirect methods (e.g., evaluating relationship between investment and liquidity) have a number of limitations. Claessens and Tzioumis (2006) provide a survey of measuring firm credit constraints.

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ing competitive lending rates. Finally, credit bureaus increase the cost of defaulting to borrowers by excluding defaulting borrowers from accessing credit markets in the future, thereby serving as a disciplinary device to encourage repayment.

Credit registries exist in many countries, and numerous empirical studies have demonstrated that it is an important factor in credit availability (Pagano and Jappelli 1993; Jappelli and Pagano 2002; Djankov et al. 2007). Djankov et al. (2007) significantly expand the data on credit institutions, gathering data in 133 countries over 25 years (representing every economy with a population over 1.5 million, except countries in civil conflict or inactive members of the World Bank). The presence of a credit bureau is a dichotomous variable, indicating if a private credit bureau operates in the country. According to Djankov et al. (2007, 303), "A private bureau is defined as a private commercial firm or nonprofit organization that maintains a database on the standing of borrowers in the financial system, and its primary role is to facilitate exchange of information amongst banks and financial institutions." It is the primary role of the exchange of information between lenders that distinguishes credit registries from other credit reporting agencies that collect and analyze credit information.[23] As a recent example, Kredi Kayit Bürosu was formed in 1996 by 11 Turkish banks, with a technical partnership with Experian, resulting in the first and only credit bureau in Turkey that became fully operational in 1999.[24]

While the establishment of a credit bureau registry is not the only factor relevant to credit market imperfections, the proxy is particularly well suited for this study for several reasons. First, as discussed, it has a solid foundation in economic theory and strong empirical support. For example, Djankov et al. (2007) demonstrate that credit bureau registries explain nearly 60% of the variation in private credit (as a percentage of GDP), which is more than other factors emphasized as important determinants of the development of credit markets. Second, data are available for all countries and years in which data on pollution concentrations are available, and there is significant variation over time, which is essential for this study because fixed effects is used to account for unobserved heterogeneity. Alter-

23. For example, the first organized agency to provide credit information to subscribers in the United States was the Mercantile Agency, founded in 1841 by Lewis Tappan, which would later become Dun & Bradstreet (Madison 2012). However, the primary role of this, and many other similar agencies, was not to facilitate the exchange of information between lenders.

24. The delay in operations in this instance was due to prolonged negotiations with data providers. The variable is based on the year in which operations began. The partnership with Experian is typical as credit bureaus are typically owned by or affiliated with large international firms (Experian, Equinox, and TransUnion own, or are affiliated with, half of the bureaus in the sample of 129 countries).

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native measures of credit market imperfections, such as the legal rights of creditors against defaulting debtors, have also been found to be important determinants of credit market imperfections (La Porta et al. 1997, 1998) but are not suitable proxies because they are "remarkably stable over time" (Djankov et al. 2007, 299). Finally, as demonstrated below, credit reforms facilitate an expansion of lending, and not the other way around, and credit reforms are not strongly correlated with potential time-varying confounding factors.

2.1.3. Primary Controls

Because the empirical analysis uses fixed effects, additional covariates are included to account for time-varying determinants of pollution emissions. Unless indicated otherwise, all variables are from the World Bank's WDI.[25] First, to account for scale effects, I use GDP per square kilometer in constant 2005 US dollars. Second, I use several proxies to account for the composition of inputs. Human capital is accounted for using measures of education and health, including the average number of years of education of the working population (over the age of 15) by Barro and Lee (2013) and life expectancy at birth. Physical capital per unit of GDP is from Amadou (2011) and measured using the perpetual-inventory method and initial capital stock computed as in Harberger (1978).[26]

Accounting for environmental policies is less straightforward as data on regulations are unavailable for developing countries and, when available, are difficult to compare across countries.[27] Consequently, the empirical analysis controls for several proxies for environmental regulations, while the robustness checks control for a rich set of variables related to environmental policy. First, if environmental quality is a normal good, increases in disposable income should increase demand for environmental quality and in turn promote stricter environmental regulations.[28] Second, while income per capita bears on individual demand for environmental quality, aggregate demand depends on the number of individuals exposed.[29] Toward this end, this pa-

25. http://data.worldbank.org.

26. Table B2 in appendix B reports summary statistics.

27. Brunel and Levinson (2013, 28) review various approaches to measuring the stringency of environmental regulations, concluding that "regulations are too complex and dissimilar across countries to create consistent measures of stringency, except in narrow case studies that are not generalizable."

28. For example, Dasgupta et al. (2001) report that GDP per capita explains 76% of the cross-sectional variation in a composite environmental indicator. The relationship between income and environmental quality is the subject of the extensive environmental Kuznets curve literature (see, e.g., Copeland and Taylor [2004] for a survey of the literature).

29. At the same time, it might be argued that the assimilative capacity of the environment would be inversely related to population density, implying that countries with lower population density would have lower costs of providing environmental quality.

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per uses consumption per capita and population density as proxies for environmental regulations.

It should be highlighted that the control variables are at best imperfect proxies for the variables of interest. As a consequence, to the extent that credit reforms are correlated with time-varying factors related to pollution emissions and measurement error in the controls, the estimated impact of credit reforms will also be biased. This paper also employs a rich set of additional controls, including measures of political institutions (e.g., democracy), governance (e.g., corruption), globalization (e.g., trade), and international environmental agreements; and the lead content of gasoline and labor productivity, to assess the robustness of results to confounding factors (sec. 2.4).

2.2. Analysis of Credit Bureau Reforms

This section corroborates that the establishment of a credit bureau increases lending among the countries in the sample of analysis. Specifically, I analyze the effect of establishing a credit bureau on private credit, defined as the ratio of credit from financial institutions to the private sector to GDP, from the International Monetary Fund (International Financial Statistics). For transparency, I use a nonparametric approach to assess the impact of the establishment of a credit bureau on private credit. Figure 3 plots average "normalized" private credit for countries experiencing credit reforms ("treated" countries) and the same average for countries not experiencing credit reforms ("control" countries) during the corresponding period.[30]

Figure 3 demonstrates that among both treated and control countries, private credit grew at roughly the same rate during the pre-reform period, typically not exceeding 1% annual growth. Moreover, treated countries experienced rapid growth in private credit following the establishment of a credit bureau, increasing nearly 20% over the subsequent 3 years. Concurrently, control countries experienced approximately the same rate of growth of private credit as in the pre-reform years. It should be highlighted that the effect of credit reforms is to increase the rate of growth, rather than the level, of private credit.[31] The observation that both treated and control countries experienced similar private credit growth in pre-reform years, but private credit grew more in treated countries in the post-reform years, suggests that the credit reforms are driving private credit growth, rather than the other way around.

While it is reassuring that credit reforms promote lending, it would also be reassuring if credit reforms are not correlated with factors that are related to environmental performance. For example, credit reforms might be associated with trans-

30. The details of the procedure are provided in appendix A2.

31. Similarly, Djankov et al. (2007) document that credit registries are increasing in importance over time.

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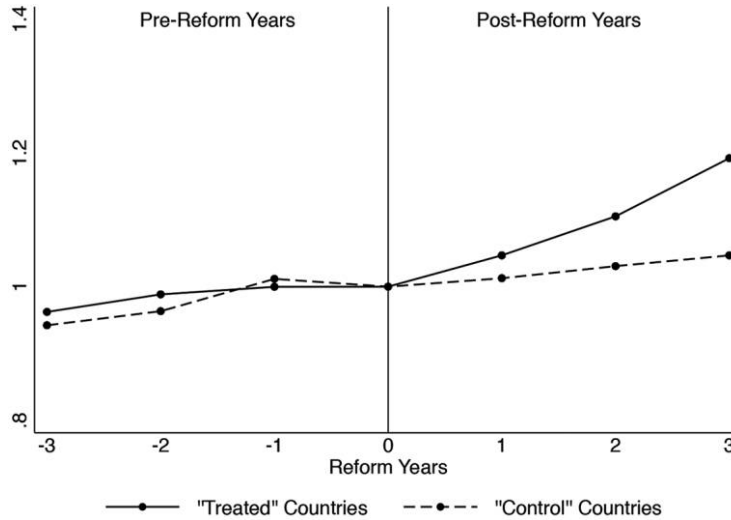

Figure 3. The effect of credit reforms on various outcomes. Averages are calculated for all countries experiencing credit reforms (treated countries) and countries not experiencing credit reforms in corresponding years (control countries). See appendix A2 for more details.

formations in political institutions or governance, which have been demonstrated to be related to environmental policy and performance.[32] Toward this end, an index of democracy and governance (government honesty) is plotted using a similar approach as above.[33]

Figure B1 in appendix B (figs. B1–B3 available online) demonstrates that trends in political institutions and governance are similar for both treated and control countries in both pre- and post-reform periods. The only exception is that government honesty is slightly higher in treated countries 2 and 3 years after the reform, though there does not appear to be any systematic trend (upward or downward) for treated and control countries. Figure B1 also demonstrates that trends in GDP per capita and trade (share of imports and exports in GDP) are remarkably similar for treated and

32. For example, Congleton (1992) and Fredriksson et al. (2005) find that democracy raises environmental policy stringency, while López and Mitra (2000) and Damania et al. (2003) find that corruption reduces stringency.

33. The measure of democracy is "polity score" from the Center for Systematic Peace Polity IV database (http://systemicpeace.org), while the measure of governance is "government honesty" from the Political Risk Services International Country Risk Guide (http://prsgroup.com).

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control countries. While these are only a few outcomes, these observations also provide indirect evidence that credit reforms are uncorrelated with unobservable confounding factors to the extent that unobservable factors are correlated with these outcomes. The determinants of credit reforms are explored more systemically and extensively in section 2.4.

Figure B2 in appendix B provides a glimpse of the impact of establishing a credit bureau on sulfur dioxide and lead concentrations. The approach is similar to the one above, except the analysis is at the site level. Moreover, a log transformation is applied to both pollutants because concentrations are highly skewed, which is consistent with the dependent variable used in the subsequent analysis. Figure B2 indicates that in pre-reform years sulfur dioxide declined, while lead remained approximately constant. Moreover, pre-reform trends for treated and control sites do not appear to be systematically different, reducing the concern of preexisting trends in pollution concentrations. In post-reform years, sulfur dioxide declines more rapidly in treated sites, though only after the second year. Similarly, lead declines in treated sites and is mostly constant in control sites.

2.3. Panel Regression Analysis

The panel regression analysis assesses the effect of credit reforms on pollution emissions using the establishment of a credit bureau. For every country, let t^{est} represent the year in which a credit bureau was established, and $1\{t \geq t^{est}\}$ represent an indicator variable if the country has a credit bureau at time t. The baseline regression model takes the form

$$\text{Pollution}_{sct} = \sum_{j=0}^{J} \psi_j 1\{t \geq t^{est}\} \times (t - t^{est})^j + \chi'\Psi + u_t + u_s + \varepsilon_{sct}, \tag{24}$$

where s indexes sites, c indexes countries, and t indexes years. The dependent variable is log SO_2 or log lead concentration.

Because credit reforms increase the rate of growth, rather than the level, of private credit (see fig. 3), the number of years elapsed since the credit reform $(t - t^{est})$ is interacted with an indicator variable if the country has a credit bureau. The coefficient ψ_0 represents the percentage change in pollution associated with the credit reform. The coefficient ψ_1 captures the additional effect of the credit reform in subsequent years, while ψ_2 captures diminishing effects (or second-order effects in general) of the credit reform in subsequent years, and so on. Including the quadratic term (and higher polynomials) allows for the effect of the credit reforms far in the past to have little or no effect on pollution. The baseline analysis includes multiple values of $J = 0$, 1, and 2.

The vector χ'_{ct} includes country-level covariates, including GDP per square kilometer (GDP/Area), household expenditures per capita (Consumption/Population),

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average number of years of education (Years Education), life expectancy at birth (Life Expectancy), and physical capital per GDP (Physical Capital/GDP).[34]

The year-specific effect (u_t) and site-specific effect (u_s) indicate that the model allows for the intercept to vary over sites and over time, thereby accounting for site unobserved effects and aggregate time effects. Because the panel is short, estimation is by fixed effects where the site-specific effect (u_s) is eliminated using a within transformation. Fixed effects produces the DID estimate of the (treatment) effect of credit reforms, which accounts for time effects and allows for controls. The DID estimator requires strict exogeneity, which implies that ε_{sct} is uncorrelated with all explanatory variables in all time periods.

Strict exogeneity therefore implies that credit reforms should be uncorrelated with the error term in all periods. The primary threats to consistency of the credit reform coefficients are therefore correlation with time-varying confounding factors, such as measurement error in the explanatory variables and omitted variables. For example, a reduction in government corruption might increase the stringency of environmental regulations and the likelihood of credit reforms. Moreover, because the panel is highly unbalanced, correlation between credit reforms and the probability of being in the sample (selection) is another potential confounding factor. Similarly, a reduction in government corruption might also increase the likelihood of reporting pollution data.

Because it is likely that the error term is correlated over time for a given site, I use cluster-robust standard errors that cluster on site.[35] While the primary variable of interest is at the country level, using standard errors that cluster on countries would result in upward biased standard errors as there is a problem of "few" clusters (Cameron and Miller 2015). Because the bias with few clusters can be severe, especially with unbalanced panels, the baseline model does not cluster at the country level. Even with the more conservative standard errors that are clustered at the country level, however, the credit reform coefficients remain statistically significant.[36]

2.3.1. Panel Regression Results

Table 1 reports the panel regression results, using log SO_2 (cols. 1–3) and lead (cols. 4–6) as dependent variables. Columns 1 and 4 use a dummy for credit bureau established, columns 2 and 5 add (to the baseline model) the number of years elapsed

34. All covariates are contemporaneous and in logs. The results are nearly identical using 1-year lagged covariates and quadratic covariates (available upon request). See section 2.1.3 for a detailed discussion of the variables.

35. For example, it is possible that a large shock in pollution emissions might increase concentration over several years.

36. Appendix A2 provides a more detailed discussion of the problem and reports the standard errors.

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Table 1. Determinants of SO_2 and Lead Concentrations Using Site Fixed Effects

	SO_2			Lead		
	(1)	(2)	(3)	(4)	(5)	(6)
Bureau established (BE)	−.0078	.0020	−.0192	−.7469†	−.7303†	−.5860†
$1\{t \geq t^{est}\}$	(.0687)	(.0633)	(.0665)	(.1777)	(.1826)	(.1613)
No. of years since BE		−.0549†	−.0468**		−.0588**	−.1047†
$1\{t \geq t^{est}\} \times (t - t^{est})$		(.0165)	(.0221)		(.0272)	(.0262)
No. of years since BE2			−.0001			.0016†
$1\{t \geq t^{est}\} \times (t - t^{est})^2$			(.0002)			(.0004)
GDP/Area	1.6067†	.8414**	.8168**	1.5887†	.8528	1.2468**
	(.3317)	(.3436)	(.3475)	(.5054)	(.5274)	(.5027)
Consumption/ Population	−1.8264†	−1.4659†	−1.4578†	.7390	.7978	1.6101**
	(.3668)	(.3572)	(.3604)	(.7091)	(.6891)	(.7208)
Physical Capital/GDP	−.8262**	−1.0575†	−1.0852†	.4047	.1378	.5976
	(.3316)	(.3252)	(.3287)	(.5194)	(.4919)	(.4894)
Years Education	−1.2671**	−1.0642	−.7988	−.3779	−.2529	−3.7631†
	(.6399)	(.6651)	(.7911)	(.8187)	(.8018)	(.9865)
Life Expectancy	2.6826	−1.4360	−1.7244	4.1788	3.7751	9.4468
	(3.1622)	(3.0356)	(3.0781)	(8.0484)	(7.8452)	(7.4755)
Population Density	.5497	.7564	.6316	−1.8886	−2.4524	−.2097
	(1.0154)	(.8986)	(.9239)	(1.8361)	(1.7267)	(1.6751)
Adjusted R^2	.311	.329	.329	.602	.607	.624
No. of sites	298	298	298	137	137	137
No. of observations	2,157	2,157	2,157	783	783	783
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes

Note.—The dependent variable is SO_2 (cols. 1–3) and lead (cols. 4–5). The dependent variables and covariates are expressed in logs. The estimations use cluster-robust standard errors that are clustered on sites. Standard errors are in parentheses.

* $p < .10$.

** $p < .05$.

† $p < .01$.

since credit bureau established, and columns 3 and 6 add the number of years squared since the credit bureau established.[37]

Using SO_2 as a dependent variable, the credit reform dummy is statistically insignificant in all specifications. However, adding the number of years elapsed since the credit reform interacted with the credit reform dummy increases the goodness of

37. Higher order polynomials are insignificant and did not increase the goodness of fit.

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fit and suggests that credit reforms reduce pollution in subsequent years (col. 2). Adding the number of years squared interacted with the credit reform dummy does not increase the goodness of fit, and the additional polynomial term is insignificant (col. 3). Columns 2 and 3 suggest that credit reforms result in approximately a 0.25% reduction in SO_2 concentration after 5 years, a 0.5% reduction after 10 years, and a 1% reduction after 20 years (and so on).

Using lead as a dependent variable, the credit reform dummy is negative and statistically significant in all specifications. Adding the number of years elapsed since the credit reform interacted with the credit reform dummy increases the goodness of fit and suggests that credit reforms reduce pollution in subsequent years (col. 5). Adding the number of years squared interacted with the credit reform dummy increases the goodness of fit, and the additional polynomial term suggests that the negative effect is diminished in subsequent years (col. 6). Column 6 suggests that credit reforms result in an immediate reduction in lead by approximately 0.6%, a 1.1% reduction after 5 years, a 1.6% reduction after 10 years, and a 2.6% reduction after 20 years (and so on).

While the theoretical model does not predict the sign of the covariates, the results are mostly consistent with expectations. For example, an increase in the scale of production increases pollution, with most models failing to reject a unitary elasticity. An increase in household disposable income, proxied by household consumption per capita, reduces SO_2 concentration, while in most specifications it does not significantly affect lead concentration. Finally, an increase in physical capital and human capital appears to decrease pollution, although the effects are insignificant in several specifications. One interpretation of this result is that lower levels of capital intensity might be associated with more depreciated or generally less efficient physical capital, which might generate more emissions. Similarly, lower levels of human capital might be associated with higher reliance on low-skilled workers or less human-capital intensive industries.

2.3.2. Discussion of Panel Regression Results

What can explain the apparent differences between the SO_2 and lead regressions? First, it appears that the model performs better using lead as a dependent variable compared to SO_2, as evidenced by the nearly twofold difference in adjusted R-squared. On the one hand, because anthropogenic sources represent only one-third to one-half of SO_2 concentration, it is unsurprising that the model explains "only" roughly 30% of the variation in concentration. On the other hand, anthropogenic sources represent nearly all of the lead concentration in the air, and consequently the model does a better job at explaining the data.

Second, it appears that the impact of credit bureau reforms is larger in magnitude for lead than SO_2. However, the impacts are quite similar when considered as

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a percentage of anthropogenic pollution. For example, after 10 years, a credit reform should reduce SO_2 concentration by 0.5%, while it should reduce lead concentration by 1.1%. However, if 50% of SO_2 concentration is from anthropogenic sources, then credit reforms reduce the concentration attributed to human activity by 1%. Thus, the impacts on anthropogenic emissions are quite similar.

Third, what can explain the observation that the impact of credit reforms on pollution, especially SO_2 concentrations, unfolds over the course of several years? First, recall from section 2.2 that credit reforms increase in importance over time, implying that credit reforms would consequently affect pollution over time as well. Second, because firms have various technology-capital vintages at the time of the credit reform, the short-run effect of credit reforms on technology upgrading will only be reflected among firms with sufficiently depreciated capital or new entrants.[38] Moreover, because new entrants typically lack credit histories, the effect of credit reforms on reducing credit constraints would only take place over the course of several years among new entrants.[39]

Finally, because the covariates might be endogenously related to credit reforms, the credit reform coefficients might reflect only partial effects. In particular, credit reforms might promote investment in physical capital, which in turn is related to pollution. Because physical capital is negatively related to SO_2, the credit reform coefficients would therefore understate the total effect of credit reforms on pollution. In assessing the extent that partial and total effects differ, indirect evidence suggests a limited degree of divergence as (i) credit reforms are not significantly related to within-country variation in physical capital, and (ii) using various lags (1–3 years) of physical capital does not significantly change the credit reform coefficients.[40] Simi-

38. In the model, technology decisions are made once each period (at the beginning), implying that in general there would be a delay in the impact of a change in credit constraints on technology decisions. With uniform (identical) technology-capital vintages, this impact would take place all at once, while with uneven technology-capital vintages, the impact would be spread out over time.

39. One reason that the effect of credit reforms on SO_2 concentrations might be particularly delayed is therefore because electricity generation, the primary source of SO_2 emissions, entails significant capital investment that depreciates over many years, and relatively few new entrants. For example, upgrading to a combined-cycle natural gas plant typically entails 2 years of construction and one additional year before achieving full capacity, while the plant life is roughly 30 years or more.

40. To be precise, the coefficient of establishing a credit bureau in a fixed effects regression similar to the baseline model, but using (log) Physical Capital/GDP as a dependent variable, is 0.006 (SE = 0.036). Moreover, using various lags of physical capital does not change the credit reform coefficients by more than one standard error.

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larly, credit reforms might increase income, and in turn more stringent environmental regulations, which would imply that the credit reform coefficients underestimate the total effect of credit reforms. Because the link is indirect, and credit constraints impact economic growth to a limited extent (see fig. B1), the degree of divergence is likely modest.

2.4. Confounding Factors

Time-varying omitted variables and nonrandom measurement error correlated with credit reforms and pollution (i.e., confounding factors) are potential sources of bias to the credit reform coefficients. The aim of this section is to assess the extent that the credit reform coefficients are robust with respect to potential confounding factors. Toward this end, this section explores whether credit reforms are correlated with variables that previous studies find to be related to environmental performance. Second, this paper employs these variables as additional controls to assess the robustness of the credit reform coefficients to the inclusion of additional controls. The aim is to rule out the possibility that the results are due to confounding observable factors, such as lead phase-out programs. Moreover, by selecting a range of observable factors that are correlated with unobservable confounding factors, it is possible to assess the extent that credit reform coefficients are robust to unobservable confounding factors (Altonji, Elder, and Taber 2005).

The selection of controls is guided by previous studies, with the constraint that variables need to be annual and cover the panel of countries. The additional controls comprise four sets of variables, including measures of political institutions, governance, globalization, and international environmental agreements as well as the lead content of gasoline and labor productivity.[41] These covariates represent a wide range of factors related both directly and indirectly to pollution emissions. For example, studies demonstrate that ratification of international environmental agreements and lead phase-out programs are correlated with broader environmental policies (Congleton 1992; Damania, Fredriksson, and List 2003; Cole, Elliott, and Fredriksson 2006). Moreover, I account for labor productivity because several factors related to productivity, such as political institutions, governance, and globalization (Hall and Jones 1999; Acemoglu, Johnson, and Robinson 2001; Rodrik, Subramanian, and Trebbi 2004), are

41. Political institutions includes measures of democracy, autocracy, regime durability, executive recruitment, executive constraints, political competition, legislative fractionalization, executive and legislative competitiveness, and executive party orientation; governance includes government honesty, bureaucratic quality, law and order, and social conditions; globalization includes trade, FDI, economic integration, political integration, and social integration; and international environmental agreements includes all treaties related to air pollution.

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also important determinants of environmental policy. To conserve space, the additional controls are discussed in more detail in appendix A2.

To assess the extent that credit reforms are correlated with potential confounding factors, a dummy variable indicating the establishment of a credit bureau is used as a dependent variable. Similar to the baseline model (eq. [24]), I use a linear fixed effects model with country and year fixed effects and the same set of control variables. Table B3 in appendix B (tables B1–B5 available online) demonstrates that credit reforms are not strongly correlated with potential confounding factors, although there are a few exceptions. The joint significance of each set of additional controls is tested using an *F*-test, indicating that political institutions, globalization, international agreements, and productivity are not jointly significant. Governance and the lead content of gasoline, however, are jointly significant ($p = .01$ and $p = .09$). Finally, the joint significance of all of the added controls is significant at the 1% level.

What conclusions can be inferred from the results? First, credit reforms do not appear to be strongly correlated with potential confounding factors, suggesting a limited degree of bias. Among the added controls that are jointly significant, credit reforms are positively correlated with government honesty and the lead content of gasoline, although sample selection bias is a concern with the latter. If government honesty is positively correlated with unobservable factors that reduce pollution, such as the stringency of environmental regulations, then the credit reform coefficient would be biased upward, while on the other hand, if countries with more lead in gasoline have similarly lax environmental regulations, then the credit reform coefficients would be biased downward.

Next, this paper employs these variables as additional controls in the primary model. Table B4 in appendix B reports the credit reform coefficients for SO_2 (panel A) and lead (panel B) samples using the baseline specifications from table 1 (cols. 2 and 5). Table B4 reports the baseline model without additional controls (col. 1), with sets of additional controls added sequentially (cols. 2–7), and with all sets of additional controls added simultaneously (col. 8). Note that column 8 does not include lead in gasoline as the sample is markedly reduced. Table B4 demonstrates that the credit reform coefficients are highly consistent with the baseline model across all specifications. The only exception is specification (6), which accounts for lead in gasoline; however, the difference can be entirely attributed to differences in the sample.[42] The joint *F*-test (*p*-value reported) in table B4 also demonstrates that political institutions, international agreements, and lead in gasoline, are significant deter-

42. That is, the credit reform coefficients are not statistically different (and in fact are quite similar) using the restricted sample in specification (6), without controlling for lead in gasoline (see the table notes).

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minants of SO_2, while political institutions, globalization, and international agreements are significant determinants of lead. Because table B3 demonstrated that governance and lead in gasoline are the only factors correlated with credit reforms, it follows that lead in gasoline in the SO_2 sample is the only potential confounding factor. As argued above, this should generate a downward bias in the credit reform coefficients, which is consistent with column 6.

Sample selection bias is another potential concern as not reporting pollution data might be nonrandom. More importantly, the probability of appearing in the sample (selection) might be correlated with the idiosyncratic errors and credit reforms, thereby biasing the credit reform coefficients. Using a simple test by Wooldridge (1995) demonstrates that selection might be nonrandom in the more restricted lead sample, but not the SO_2 sample, suggesting that the credit reform coefficients might be biased in the lead sample. To correct for sample selection bias in the lead sample, I use the Wooldridge (1995) selection correction procedure, which demonstrates that the credit reform coefficients are similar in magnitude and significance as the baseline results. That is, the point estimate of the establishment of a credit bureau is −0.55 and is significant at the 1% level, while the point estimate of the number of years established is −0.06 and is significant at the 5% level. To conserve space, the details of the test and correction procedure are provided in appendix A2.

2.5. Differential Effect of Credit Reforms

Legal origin has been demonstrated as an important determinant of investor protection and, in turn, the development of credit markets (La Porta et al. 1997, 1998). In particular, among English legal origin countries, credit market institutions (legal rules) provided strong protection to shareholders and creditors, while among civil law countries, especially French civil law countries, credit market institutions provided less protection to shareholders and creditors.[43] As a result, common law countries have markedly higher creditor rights than civil law countries and, consequently, have larger and wider credit markets. Djankov et al. (2007) demonstrate that the establishment of a credit bureau is particularly beneficial to widening credit markets in French legal origin countries. One explanation is that the provision of information conferred from credit bureaus might serve as a substitute to creditor rights.

43. The English "common law system" is a legal system affording precedence to the adherence of previous "precedencial" decisions, which originated in England and spread throughout the colonies of the British Empire, including Australia, Canada, India, and the United States. The French "civil law system" is a legal system relying on codified statutes or regulations, which originated in France and was propagated to the countries conquered by Napoleon, including Spain and Portugal, and their colonies.

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Figure B3 in appendix B plots average normalized private credit for countries experiencing credit reforms in French and English legal origin countries, following the approach used in section 2.2. However, in this case, treated countries are divided into French and English legal origin countries. The effect of credit reforms on private credit is markedly greater in French legal origin countries compared to English legal origin countries. In particular, private credit growth in English and French legal origin countries is nearly identical in pre-reform years but is approximately 20 percentage points higher in French legal origin countries in post-reform years.

The aim of assessing the differential effect of credit reforms by legal origin is not only to account for heterogeneity but also to provide an additional robustness check. That is, to the extent that differences in credit reforms between French and English legal origin countries reflect differences in the extent that credit constraints are reduced (and not other differences), comparing the effect of credit reforms in French and English legal origin would eliminate any potential bias that is common across both French and English legal origin countries.

The effect of credit reforms might also depend on the level of development. For example, in the absence of credit bureau registry, developed countries might have developed alternative institutions to overcome asymmetric information, suggesting that establishment of a credit bureau would be less important in developed countries compared to developing countries. Moreover, empirical studies document that firms in developing countries are severely credit constrained with very high rates of return on capital (Banerjee and Duflo 2014). To account for differential effects by legal origin and the level of development, an indicator variable for French legal origin ("French") and an indicator for Developed are interacted with credit reforms. Developed indicates that GDP per capita evaluated at the time of the credit reform is above the median GDP per capita of all countries evaluated at the year in which they experienced a credit reform. The results are reported in table B5 in appendix B. Columns 1 and 4 account for heterogeneous effects by legal origin, columns 2 and 5 by level of development, and columns 3 and 6 account for both sources of heterogeneity.

From columns 1 and 4, the results indicate that among French legal origin countries the effect of credit reforms on pollution emissions is negative and statistically significant in both the short run and long run for both SO_2 and lead concentrations.[44] Moreover, there is a significant differential impact between French and English legal origin countries: in the long run, there is a larger (negative) effect on

44. For French legal origin countries, the short-run effect on SO_2 is calculated by adding the first and third rows ($0.099 - 0.311 = -0.211$), while the effect of additional years (long-run effect) is calculated by adding the second and fourth rows ($-0.070 + 0.030 = -0.040$), and similar for lead. The corresponding standard errors are calculated using the delta method.

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SO_2 concentration for French countries, while in both the short and long run, there is a larger effect on lead concentration.[45]

Columns 2 and 5 demonstrate that the impact of credit reforms is less acute in developed countries, although the difference is not statically significant in the SO_2 sample. Columns 3 and 6 demonstrate that the conclusions regarding differential impacts by legal origin and level of development are consistent when both sources of heterogeneity are accounted for.

2.6. Alternative Approach: Difference in Differences

Panel regressions are subject to criticism as they often omit relevant time-varying factors or because independent variables are endogenous. This section uses an alternative DID approach in which pollution concentrations are analyzed around credit reforms. Specifically, I restrict the sample to countries experiencing a credit reform during the sample and collapse the panel into a pre- and post-period following the approach recommended by Bertrand et al. (2004). Differences in pollution after and before credit reforms for countries experiencing credit reforms within a pre- and post-period time frame (treated) are then compared to the same difference for countries not experiencing a credit reform within that time frame (control).

The primary advantage of the alternative DID approach is its simplicity and potential to overcome the endogeneity problems that arise in panel regressions. The alternative DID approach relies on the assumption that there is no interaction between time effects and credit reforms conditional on site and year fixed effects. In other words, identification requires that, in the absence of credit reforms, trends in pollution concentrations for treated and control sites would not be systematically different (parallel paths). For example, if changes in the stringency of environmental regulations are correlated with the timing of credit reforms, then the parallel paths assumption would be violated.

One common approach to shed light on the parallel paths assumption is to compare trends in the outcomes of interest in years prior to the reform, as well as trends in potential confounding factors. As mentioned, figure B2 supports this assumption as pre-reform pollution trends are similar for both treated and control sites, and trends in potential confounding factors are also similar.

Following Bertrand et al. (2004), I run the following cross-sectional regression

$$\Delta \text{Pollution}_s^T = \zeta_0 + \zeta_1 \Delta \text{Pollution}^C + \varepsilon_s, \tag{25}$$

45. For example, after 5 years, credit reforms reduce SO_2 concentration by 0.25% in English legal origin countries, whereas they reduce concentration by 0.42% in French legal origin countries. Similarly, credit reforms reduce lead concentration by 0.68% in English legal origin countries, whereas they reduce concentration by 2% in French legal origin countries.

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where $\Delta\text{Pollution}_s^T$ is the difference in average levels of pollution after and before the credit reform for site s, and $\Delta\text{Pollution}^C$ is the same difference averaged over all sites not experiencing a credit reform (countries experiencing a credit reform during the sample of analysis, but not during the pre- or post-reform years).[46] The constant term ζ_0 therefore represents the average effect of the credit reform, controlling for systematic changes in nonreforming countries (DID estimate).

An inherent trade-off exists in the selection of the optimal number of pre- and post-reform years. On the one hand, using few years increases the sample size and reduces the potential for omitted variable bias. On the other hand, it increases the probability of Type II error whenever the effect is delayed or incremental. The sample of sites experiencing credit reforms, especially for lead, is small and consequently the range of years to explore is significantly constrained.

Table 2 reports the results for SO_2 and lead pollution concentration. The DID estimate of the impact of the credit reform for SO_2 is insignificant using one and two pre-/post-reform years, while it is negative and significant using three pre-/post-reform years. This is consistent with the delayed effect of credit reforms on SO_2 concentrations observed in the panel regressions and figure B2. The DID estimate of the impact of the credit reform for lead is negative and significant using one and two pre-/post-reform years. Moreover, the cumulative effect over 2 years is larger than the effect over the first year.[47]

3. CONCLUSION

This paper presents a general equilibrium model to analyze the effect of credit constraints on the environment, demonstrating that the effect consists of technique, scale, upgrading-composition, and market-size effects. Due to the confounding nature of the effects, the net effect of credit constraints on aggregate pollution emissions is ambiguous. The model derives relatively simple necessary and sufficient conditions for credit constraints to increase or decrease aggregate pollution emissions. Inspection of the relevant parameter space indicates that reducing credit constraints plausibility leads to a net reduction in pollution emissions.

Motivated by the ambiguity pointed out in the theoretical model, this paper exploits credit market reforms as exogenous variation in credit constraints. Using panel regression analysis to produce a DID estimate of credit reforms indicates that the establishment of a credit bureau reduces sulphur dioxide and lead concentrations by 0.25% and 1.1% after 5 years, respectively. However, because anthropogenic sources represent between one-third and one-half of sulphur dioxide, the effect corresponds to

46. The details of the procedure are provided in appendix A2.

47. Additional pre-/post-reform years results are insignificant and are not reported due to small sample size.

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Table 2. Difference in Differences Estimates of SO_2 and Lead Concentrations

	SO_2			Lead	
	(1)	(2)	(3)	(4)	(5)
Constant (DID estimate)	.0640	−.0356	−1.2045**	−1.1621†	−1.8871†
	(.0504)	(.0665)	(.4036)	(.3767)	(.3352)
No. of pre- and post-reform years	1	2	3	1	2
Adjusted R^2	.217	.398	.189	.186	.625
No. of observations	36	23	12	14	13

Note.—The dependent variable is (log) SO_2 (cols. 1–3) and lead (cols. 4–5). Standard errors are in parentheses.

* $p < .10$.

** $p < .05$.

† $p < .01$.

between a 0.5% and 0.75% reduction in sulphur dioxide from anthropogenic sources. Moreover, the effect of credit reforms is significantly more pronounced in French legal origin countries, where credit reforms are more important for credit intermediation. For example, in French legal origin countries, credit reforms reduce sulphur dioxide concentration by 0.42% and lead concentration by 2% after 5 years. The results are also consistent using an alternative DID approach, wherein the time series data are collapsed and pollution emissions are analyzed around credit reforms.

This paper allays the concern that the trade-off between economic development and environmental quality is inevitable. However, the source of economic development matters—growth achieved through credit intermediation and average productivity growth might reduce pollution, while growth achieved through capital accumulation might increase pollution. While many countries have established credit bureau registries, significant variation in other credit market institutions persists across countries, such as the legal rights of creditors against defaulting debtors, which have a significant positive role in credit intermediation (Djankov et al. 2007). Increasing creditor rights therefore represents a potential "win-win" policy reform in terms of promoting economic growth and reducing environmental damages.

This paper represents a first step in understanding the role of credit constraints in environmental performance; however, it should not be the last. An important caveat is that the analysis is relevant to production-generated pollution emissions (smokestack pollution) and might not apply to consumption-generated pollution (tailpipe). While it is plausible that reducing household credit constraints might reduce tailpipe pollution, extrapolating the results to tailpipe pollution should not be taken for granted. Toward this end, future research might develop a conceptual framework for assess-

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ing the effect of household credit constraints on tailpipe pollution and investigate the relationship using consumption-generated pollutants. Future research might also use firm-level data to shed light on the relevant microeconomic parameters. Moreover, identification of the various parameters described in the model would provide greater insights into the role of credit constraints on pollution emissions.

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